

1 Linear Regression

Models

Bias of $\hat{b}_f$ $b_f = E[\hat{y}_i] - y_i$

Bias-Var Trade-off $E[(y_0 - \hat{y}_0)^2] = \text{var}(\hat{y}_0) + \text{Bias}(\hat{y}_0)^2 + \text{var}(\epsilon)$

$$\text{TSS} = \sum_{i=1}^n (y_i - \bar{y})^2 \quad \text{RSS} = \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

$$\text{MSE(LOSS)} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2 \quad \mathbf{F} = \frac{(\text{TSS} - \text{RSS})/p}{\text{RSS}/(n-p-1)}$$

$$R^2 = 1 - \frac{\text{RSS}}{\text{TSS}} \quad \text{RMSE} = \sqrt{\frac{\text{RSS}}{n-p-1}}$$

2 Classification

ROC

$$\text{Sensitivity(Recall)} = \frac{TP}{TP + FN}$$

$$\text{Specificity} = \frac{TN}{TN + FP}$$

$$\text{Precision} = \frac{TP}{TP + FP}$$

Box-Cox Transformation

$$y = \begin{cases} \frac{x^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \log(x), & \lambda = 0 \end{cases}$$

Naive Bayes

$$P(Y=k|X=x) = \frac{1}{Z} P(Y=k) \prod_{i=1}^p P(X_i = x_i | Y=k)$$

3 Model Selection

Subset Selection

Given model M_0 for null model, M_p for full model, and M_k for k predictors,

- **Best Subset Selection:** For $k=1, 2, \dots, p$, fit all $\binom{p}{k}$ models and choose the best one.
- **Forward Stepwise Selection:** Start with M_0 and add one variable at a time.
- **Backwards Stepwise Selection:** Start with M_p and remove one variable at a time.

Ridge and Lasso

Ridge Regression

$$\min \left\{ \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \hat{\beta}_j^2 \right\}$$

Lasso Regression

$$\min \left\{ \sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda |\hat{\beta}_j| \right\}$$

$$\text{RSS}(\lambda) = (y - X\hat{\beta})^T (y - X\hat{\beta}) + \lambda \beta^T \beta, \text{ solving gives } \hat{\beta} = (X^T X + \lambda I)^{-1} X^T y$$

4 Polynomial Regression and Spline

Polynomial Regression

$$y_i = \beta_0 + \beta_1 x_i + \beta_2 x_i^2 + \dots + \beta_d x_i^d + \epsilon_i$$

Giving K knots, we have $K+1$ intervals.

$$C_0(X) + C_1(X) + \dots + C_{K-1}(X) + C_K(X) = 1.$$

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \dots + \beta_K C_K(x_i) + \epsilon_i$$

- Freedom of Cubic Spline: $K+4$ degrees of freedom.
- A degree- d spline is a piecewise degree- d polynomial with continuity in derivatives up to degree $d-1$.
- To fit a cubic spline with K knots, we fit a linear regression with an intercept and the $K+3$ predictors: $x, x^2, x^3, h_1(x, \xi_1), h_2(x, \xi_2), \dots, h_K(x, \xi_K)$.
- A natural cubic spline with K knots uses K degrees of freedom

Smoothing Splines

We can try and ensure that $g(x)$ is smooth by choosing $g(x)$ that minimizes $\sum_{i=1}^n (y_i - g(x_i))^2 + \lambda \int g''(x)^2 dx$.

Effective Degrees of Freedom: $df_\lambda = \sum_{i=1}^n [S_\lambda]_{ii}$, the trace of S_λ .

MLR

Linear Model: $Y = X\beta + e$ with $E[\epsilon] = 0$ and $\text{var}(\epsilon) = \sigma^2 I_{n \times n}$

Estimator of β : $Q(\beta) = \|y - X\beta\|^2$

Estimator of variance(σ^2): $s_e^2 = \frac{1}{n-p-1} \|y - X\hat{\beta}\|^2$

Slope: $\hat{\beta} = (X^T X)^{-1} X^T y$

Simple LR: $\hat{\beta}_1 = \sum_{i=1}^n \frac{x_i - \bar{x}}{\sum_{i=1}^n (x_i - \bar{x})^2} y_i \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$

$$\text{FPR} = \frac{FP}{FP + TN}$$

$$\text{TPR} = \frac{TP}{TP + FN}$$

$$\text{F1-Score} = \frac{2TP}{2TP + FP + FN}$$

LDA & QDA

$$\text{LDA: } \delta_k(x) = \frac{x\mu_k}{\sigma^2} - \frac{\mu_k^2}{2\sigma^2} + \log(\pi_k)$$

$$\text{LDA (p > 1): } \delta_k(x) = x^T \Sigma^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma^{-1} \mu_k + \log(\pi_k)$$

$$\text{QDA: } \delta_k(x) = -\frac{1}{2} x^T \Sigma_k^{-1} x + x^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \mu_k^T \Sigma_k^{-1} \mu_k - \frac{1}{2} \log|\Sigma_k| + \log(\pi_k)$$

Metrics

$$C_p = \frac{1}{n} (\text{RSS} + 2p\hat{\sigma}^2)$$

$$C'_p = \frac{\text{RSS}}{\hat{\sigma}^2} + 2p - n$$

$$\text{AIC} = 2p - 2\log(\hat{L})$$

$$\text{AICc} = \text{AIC} + \frac{2p(p+1)}{n-p-1}$$

$$\text{BIC} = \log(n)p - 2\log(\hat{L}).$$

$$\text{Adjusted } R^2 = 1 - \frac{\text{RSS}/(n-p-1)}{\text{TSS}/(n-1)}$$

LOOCV

$$\text{MSE}_i = (y_i - \hat{y}_i)^2 \quad \text{CV}_{(n)} = \frac{1}{n} \sum_{i=1}^n \text{MSE}_i,$$

$$\text{CV}_{(n)} = \frac{1}{n} \sum_{i=1}^n \left(\frac{y_i - \hat{y}_i}{1 - h_i} \right)^2, \quad h_i = \text{diag}(X(X^T X)^{-1} X^T)$$

Cubic Spline

A cubic spline with K knots,

$$y_i = \beta_0 + \beta_1 C_1(x_i) + \beta_2 C_2(x_i) + \dots + \beta_K C_K(x_i) + \epsilon_i.$$

Truncated polynomial basis functions:

$$h(x, \xi) = (x - \xi)_+^3 = \begin{cases} (x - \xi)^3, & x > \xi \\ 0, & x \leq \xi \end{cases}$$

5 KNN

KNN Regression

$$\hat{P}(Y=j|X=x_0)=\frac{1}{K}\sum_{i\in\mathcal{N}_0}I(y_i=j)$$
$$\hat{f}(x_0)=\frac{1}{K}\sum_{i\in\mathcal{N}_0}y_i$$

Nadaraya–WatsonKernel-WeightedAverage

$$\hat{f}(x_0)=\frac{\sum_{i=1}^nk_{\lambda}(x_0,x_i)y_i}{\sum_{i=1}^nk_{\lambda}(x_0,x_i)}$$

Epanechnikov quadratic kernel

$$K_{\lambda}(x_0,x)=D\left(\frac{|x-x_0|}{\lambda}\right)$$
$$D(t)=\begin{cases}\frac{3}{4}(1-t^2), & |t|<1\\0, & \text{otherwise}\end{cases}$$

6 MARS and CART

MARS

$$\hat{f}(x)=\sum_{i=1}^kc_iB_i(x)$$

where $B_i(x)$ takes one of three forms:

- an intercept,
- a hinge function $\max(0,x-c)$ or $\max(0,c-x)$
- a product of hinge functions $h_j(x)h_k(x)$ for some ξ_j and ξ_k .

GCV

$$GCV=\frac{\frac{1}{N}\sum_{i=1}^n(y_i-\hat{y}_i)^2}{\left(1-\frac{ENP}{N}\right)^2}$$

where $ENP=k+\frac{d(k-1)}{2}$ and penalty term d is usually 2 or 3.

Michaelis-Menten Model

$$f(x,(K,V_m))=\frac{V_mx}{K+x}$$

Classification Trees

$$\hat{p}_{mk}=\frac{1}{N_m}\sum_{i\in R_m}I(y_i=k),\quad k(m)=\operatorname{argmax}_k\hat{p}_{mk}$$

Misclassification rate

$$\frac{1}{N_m}\sum_{x_i\in R_m}I(y_i\neq k(m))=1-\hat{p}_{mk(m)}$$

Gini Index

$$\sum_{k=1}^K\hat{p}_{mk}(1-\hat{p}_{mk})=1-\sum_{k=1}^K\hat{p}_{mk}^2$$

Cross entropy / deviance

$$-\sum_{k=1}^K\hat{p}_{mk}\log(\hat{p}_{mk})$$

7 Random Forest

Bootstrap

Consider a random sample $S=\{X_1,X_2,...,X_n\}$, and its bootstrap resample:

$$S_b^*=\{X_{b1}^*,X_{b2}^*,...,X_{bn}^*\},$$

where

$$\bar{X}^*=\hat{\mathbb{E}}[X^*]=\frac{1}{B}\sum_{b=1}^BX_b^*$$
$$\text{var}(X^*)=\frac{1}{B-1}\sum_{b=1}^B(X_b^*-\bar{X}^*)^2$$

$$\bar{b}_T(\theta)(\mathbf{Bias})=\hat{X}^*-X$$

Bagging

Consider $\{X_1,X_2,...,X_n\}$, $X_i\sim N(\mu,\sigma^2)$. Then,

$$\text{var}(\bar{X})=\frac{\sigma^2}{n}$$
$$\text{var}(\bar{X})=\rho\sigma^2+\frac{1-\rho}{n}\sigma^2\quad\text{with correlation }\rho$$

Guidelines

- **Classification:** Default for m is $\lfloor\sqrt{p}\rfloor$ and minimum node size of 1
- **Regression:** Default for m is $\lfloor\frac{p}{3}\rfloor$ and minimum node size of 1

SVM

P-dimensional Hyperplane

$$\beta_0+\beta_1x_1+\beta_2x_2+...+\beta_px_p=0$$

Properties

$$yi(\beta_0+\beta_1x_1+\beta_2x_2+...+\beta_px_p)>0$$

Inner Product

$$\langle x_i,x_{i'}\rangle=x_i^Tx_{i'}=\sum_{j=1}^px_{ij}x_{i'j}$$

Kernel functions

Linear

$$k(x_i,x_{i'})=\sum_{j=1}^px_{ij}x_{i'j}$$

Polynomial

$$k(x_i,x_{i'})=\left(1+\sum_{j=1}^px_{ij}x_{i'j}\right)^d$$

Radial

$$k(x_i,x_{i'})=\exp\left(-\gamma\sum_{j=1}^p(x_{ij}-x_{i'j})^2\right)$$